
Algorithm 1: Minimum-weight hyperpath from a set of seed molecules

Input: R reactions; reaction r has reagent set $T(r)$ and product set $H(r)$

S seed molecules (the generation-0 species of the network)

f combining function: max for chain length, sum for reaction count

Output: d minimum cost of reaching each species

via the reaction attaining that cost

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1  $d(m) \leftarrow 0$  for  $m \in S$ ,  $d(m) \leftarrow \infty$  otherwise;    $via \leftarrow \emptyset$ ;    $F \leftarrow \emptyset$ 
2 for each  $r \in R$  do
3    $\lfloor \text{pending}(r) \leftarrow |\{x \in T(r) : d(x) = \infty\}|$ 
4    $Q \leftarrow$  min-priority queue holding each  $m \in S$  with key 0
5   for each  $r$  with  $\text{pending}(r) = 0$  do
6      $\lfloor \text{RELAX}(r)$ 
7   while  $Q$  is not empty do
8      $(k, m) \leftarrow \text{EXTRACT-MIN}(Q)$ 
9     if  $m \in F$  or  $k > d(m)$  then continue
10     $F \leftarrow F \cup \{m\}$ 
11    for each  $r$  with  $m \in T(r)$  do
12       $\lfloor \text{pending}(r) \leftarrow \text{pending}(r) - 1$ 
13       $\lfloor \text{if } \text{pending}(r) = 0$  then  $\text{RELAX}(r)$ 
14 return  $d, via$ 

15 Function  $\text{RELAX}(r)$ :
16   if  $d(x) = \infty$  for any  $x \in T(r)$  then return
17    $c \leftarrow 1 + f\{d(x) : x \in T(r)\}$                                      //  $c \leftarrow 1$  when  $T(r)$  is empty
18   for each  $p \in H(r)$  do
19     if  $p \notin F$  and  $c < d(p)$  then
20        $\lfloor \lfloor d(p) \leftarrow c$ ;    $via(p) \leftarrow r$ ;    $\text{INSERT}(Q, p, c)$ 
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